chemotropic substances. The pollen tube after travelling the whole length of the style enters into the ovary locule where it is guided towards the micropyle of the ovule by a structure called **obturator** (See Do you know). After reaching the embryo sac, a pore is formed in pollen tube wall at its apex or just behind the apex. The content of the pollen tube (two male gametes, vegetative nucleus and cytoplasm) are discharged into the synergids into which pollen tube enters. The pollen tube does not grow beyond it, in the embryo sac. The tube nucleus disorganizes.

1.5.1 Double fertilization and triple fusion

S.G. Nawaschin and L.Guignard in 1898 and 1899, observed in *Lilium* and *Fritillaria* that both the male gametes released from a male gametophyte are involved in the fertilization. They fertilize two different components of the embryo sac. Since both the male gametes are involved in fertilization, the phenomenon is called **double fertilization** and is unique to angiosperms. One of the male gametes fuses with the egg nucleus (syngamy) to form **Zygote.** (Figure 1.19)

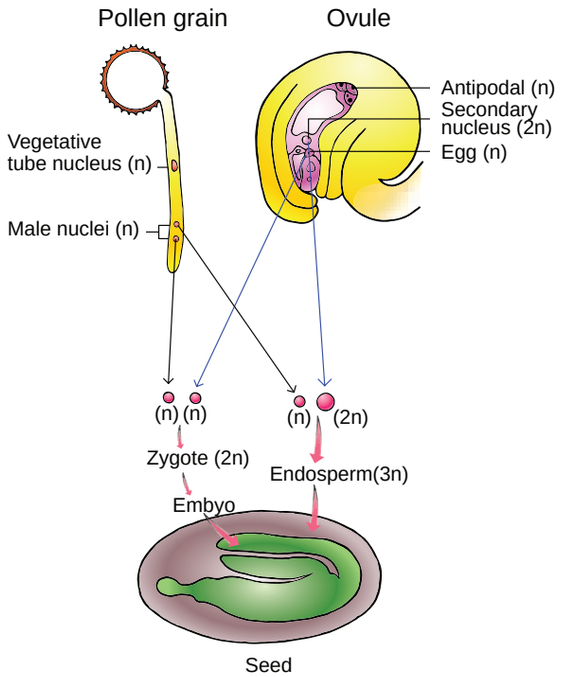


Figure 1.19 Fertilization in Angiosperms

The second gamete migrates to the central cell where it fuses with the **polar nuclei** or their

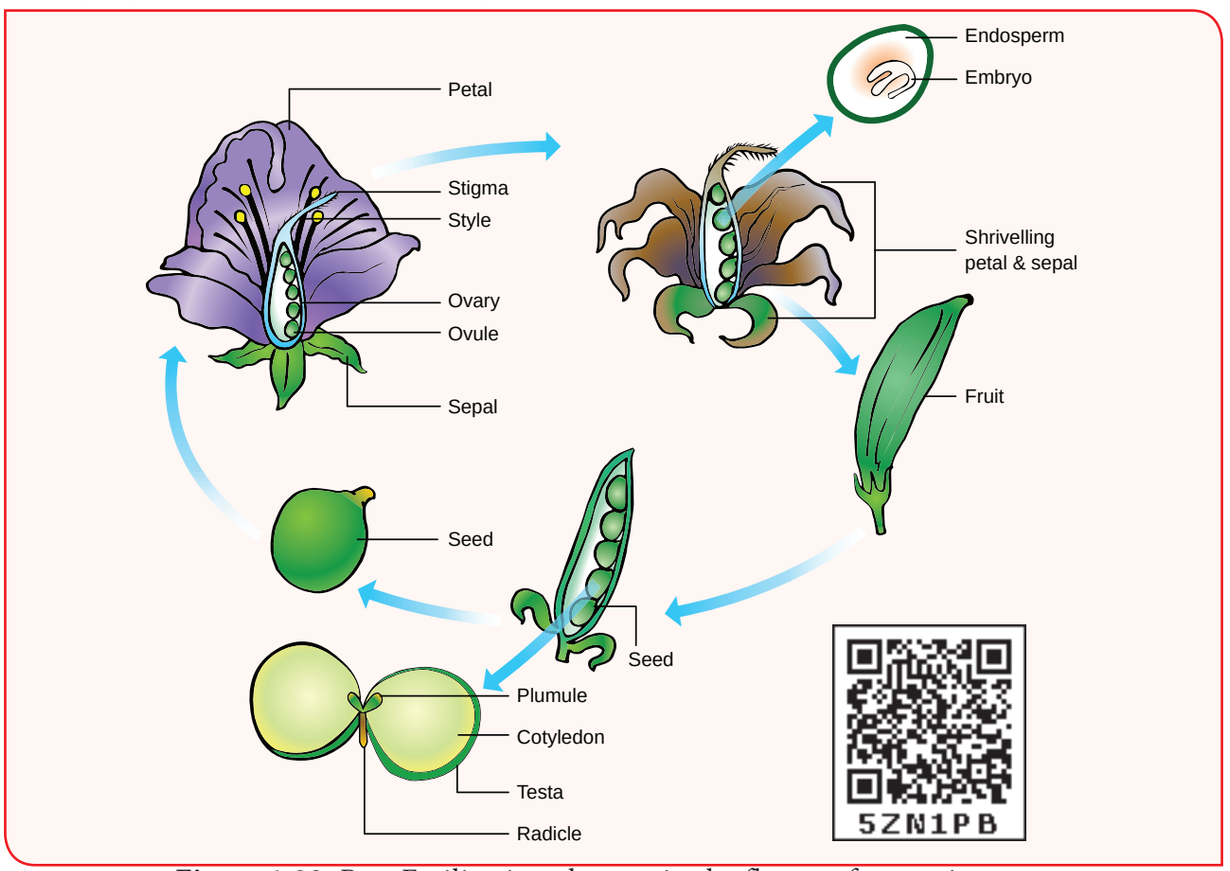


Figure 1.20 Post Ferilization changes in the flower of an angiosperm

TN_GOVT_BIOLOGY-BOTANY_XII_PAGES_001-032 CH 01.indd 20 3/14/2020 10:30:14 AM